25-08-2026
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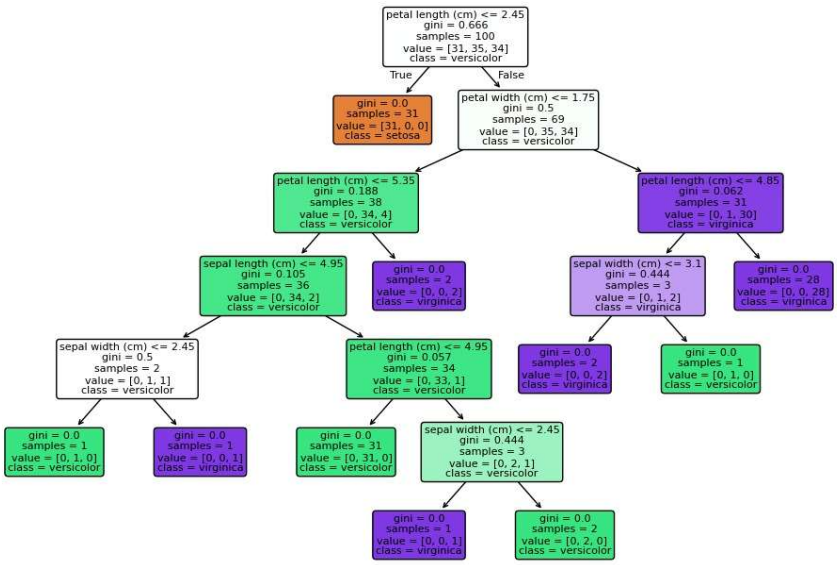
Experiment 6:

AIM: Implementation of Decision tree using sklearn and its parameter tuning

PROGRAM:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier,plot_tree
from sklearn.metrics import classification_report,accuracy_score,confusion_matrix
import matplotlib.pyplot as plt
iris=load_iris()
X=iris.data
y=iris.target
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.33,random_state=42)
treemodel=DecisionTreeClassifier(random_state=42)
treemodel.fit(X_train,y_train)
y_pred=treemodel.predict(X_test)
plt.figure(figsize=(12,8))
plot_tree(treemodel,filled=True,feature_names=iris.feature_names,
        class_names=iris.target_names,rounded=True)
plt.show()
print("Classification Report:")
print(classification_report(y_test,y_pred,target_names=iris.target_names))
print("Confusion Matrix:")
print(confusion_matrix(y_test,y_pred))
print("Accuracy Score:") print(accuracy_score(y_test,y_pred))
```

Output:



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Classification Report:

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	19
versicolor	0.94	1.00	0.97	15
virginica	1.00	0.94	0.97	16
accuracy			0.98	50
macro avg	0.98	0.98	0.98	50
weighted avg	0.98	0.98	0.98	50

Confusion Matrix: [[19 0 0]
[0 15 0]
[0 1 15]]

Accuracy Score:
0.98